

Computer Science & IT

Discrete Mathematics



Combinatorics

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



Topic

Solution of recurrence relation using method of characteristic roots



Topics to be Covered



Topic

Solution of recurrence relation using method of characteristic roots

Topic

Method of undetermined coefficient for particular solution



H.W.
Q:- Find the solution of the recurrence relation,
 $a_n = 2a_{n-1} + 2^n$ $a_0 = 1$

Soln: $a_n - 2a_{n-1} = 2^n$ ← fcn

Put $n-1 = n$, i.e. $n = n+1$

$a_{n+1} - 2a_n = 2^{n+1}$ ← Fcn

Using shift operator

$E \cdot a_n - 2a_n = 2 \cdot (2)^n$

$a_n (E-2) = \underbrace{2 \cdot 2^n}_{F(n)}$
← $\phi(E)$

$\phi(E) = E-2$

∴ Ch eqⁿ $t-2=0$

∴ Ch root $t=2$

∴ Complementary fn

CF = $C_1 \cdot (2)^n$

$a_n (E-2) = 2 \cdot (2)^n$

$a_n = \frac{2 \cdot 2^n}{(E-2)}$

P.S = Soln_{of} $\left(\frac{2 \cdot (2)^n}{(E-2)} \right)$

= 2 · Soln_{of} $\left(\frac{(2)^n}{(E-2)} \right)$ ← base 'b'=2
 2 is Ch. root with multiplicity 1

= 2 · Soln_{of} $\left(\frac{(2)^n}{(E-2)^1 \cdot 1} \right)$ → $\psi(E) = 1$
 Constant function

= 2 × $\left\{ \frac{1}{\psi(2)} \cdot \underset{m}{n} C_1 \cdot (2)^{n-1} \right\}$ ← m

P.S = $2 \cdot n \cdot (2)^{n-1} = n \cdot 2^n$

Complete Soluⁿ

$$a_n = CF + P.S.$$

$$a_n = C_1 \cdot 2^n + n \cdot 2^n$$

We know $a_0 = 1$

$$a_0 = 1 = C_1(2)^0 + 0 \cdot (2)^0$$

$$\boxed{1 = C_1}$$

$$\therefore a_n = 1 \cdot 2^n + n \cdot 2^n$$

$$\boxed{a_n = 2^n(n+1)}$$

HW Q:- Find the soln of recurrence relation,

$$a_n - 2a_{n-1} + a_{n-2} = 0$$

$$\boxed{a_0 = 1}$$
$$\boxed{a_1 = 2}$$

$$\phi(E) = E^2 - 2E + 1$$

$$F(n) = 0$$

Ch. eqn: $t^2 - 2t + 1 = 0$

$$(t-1) \cdot (t-1) = 0$$

$\therefore$ Ch roots $\underbrace{t_1 = 1, t_2 = 1}_{\text{two equal roots}}$

$\therefore$ CF = $(C_1 + C_2 n) \cdot (1)^n$
 $= (C_1 + C_2 n)$

$\therefore$ PS = 0

Hence Complete Soln

$$a_n = CF + \cancel{PS} \rightarrow 0$$

$$a_n = (C_1 + C_2 n)$$

$$a_0 = 1 = C_1 + C_2 \cdot 0 \Rightarrow \boxed{C_1 = 1}$$

$$a_1 = 2 = C_1 + C_2 \Rightarrow \boxed{C_2 = 1}$$

$$\therefore \boxed{a_n = (1 + n)}$$

H.W.

Q:- Find the solution of recurrence relation

$$a_n - 3a_{n-1} + 2a_{n-2} = 2^n$$

$$\boxed{\begin{matrix} a_0 = 1 \\ a_1 = 2 \end{matrix}}$$

$$\phi(E) = E^2 - 3E + 2$$

$$F(n) = (2)^{n+2}$$

Ch. eqn $t^2 - 3t + 2 = 0$

$$(t-2) \cdot (t-1) = 0$$

$\therefore$ Ch root $t_1 = 2$ & $t_2 = 1$

$$\begin{aligned} \therefore CF &= C_1(2)^n + C_2(1)^n \\ &= C_1 \cdot 2^n + C_2 \end{aligned}$$

$$P.S = \text{Soln of } \left(\frac{F(n)}{\phi(E)} \right)$$

$$= \text{Soln of } \left(\frac{4 \cdot (2)^n}{(E-2)(E-1)} \right)$$

$$= 4 \cdot \text{Soln of } \left(\frac{(2)^n}{(E-2)(E-1)} \right)$$

base 'b' = 2

'2' is a root with multiplicity = 1

H.W.

Q:- Find the solution of recurrence relation

$$a_n - 3a_{n-1} + 2a_{n-2} = 2^n$$

$$\begin{array}{l} a_0 = 1 \\ a_1 = 2 \end{array}$$

P.S = Soln of $\left(\frac{F(n)}{\phi(E)} \right)$

$$= \text{Soln of } \left(\frac{4 \cdot (2)^n}{(E-2)(E-1)} \right)$$

$$= 4 \cdot \text{Soln of } \left(\frac{(2)^n}{(E-2)(E-1)} \right)$$

base 'b' = 2
'2' is a root with multiplicity = 1

P.S = 4. $\left\{ \text{Soln of } \left(\frac{(2)^n}{(E-2)^1 \cdot (E-1)} \right) \right\} \rightarrow \psi(E)$

$$= 4 \cdot \left\{ \frac{1}{\psi(2)} \cdot n \cdot c_1 \cdot (2)^{n-1} \right\}$$

$$= 4 \cdot \left\{ \frac{1}{(2-1)} \cdot n \cdot (2)^{n-1} \right\}$$

$$= 2 \cdot n \cdot 2^n$$

∴ Complete Soluⁿ

$$a_n = CF + PS$$

$$= (C_1 \cdot 2^n + C_2) + n \cdot 2^{n+1}$$

We know

$$\left. \begin{aligned} a_0 = 1 &= (C_1 \cdot (2)^0 + C_2) + 0 \Rightarrow C_1 + C_2 = 1 \\ a_1 = 2 &= (C_1 \cdot (2)^1 + C_2) + 1 \cdot 2^{(1+1)} \Rightarrow 2C_1 + C_2 = -2 \end{aligned} \right\} \begin{aligned} C_1 &= -\frac{3}{4} \\ C_2 &= 4 \end{aligned}$$

$$\begin{aligned} \therefore a_n &= (-3 \cdot (2)^n + 4) + n \cdot 2^{n+1} \\ &= n \cdot 2^{n+1} - 3 \cdot 2^n + 4 \end{aligned}$$

HW Q: Find the solution of recurrence relation

$$T(3^k) = 2 \cdot T(3^{k-1}) + 1$$

$$\text{let } \boxed{T(1) = 1}$$

$$\text{let } T(3^k) = Q_k$$

$\therefore$ Eqⁿ becomes

$$\boxed{Q_k = 2 \cdot Q_{k-1} + 1}$$

↳ Now solve
the question

HW Q: Find the solution of recurrence relation

$$T(3^k) = 2 \cdot T(3^{k-1}) + 1 \quad \text{let } \boxed{T(1) = 1}$$

$$\text{let } T(3^k) = Q_k$$

∴ Eqn becomes

$$Q_k = 2 \cdot Q_{k-1} + 1$$

$$Q_k - 2Q_{k-1} = 1$$

$$\phi(E) = (E - 2)$$

$$F(n) = 1$$

$$\text{Ch eqn} \Rightarrow t - 2 = 0$$

$$\text{Ch. root} \Rightarrow t = 2$$

$$\text{CF} = C_1 \cdot (2)^k$$

$$\text{P.S.} = \text{Solun of } \left(\frac{F(n)}{\phi(E)} \right)$$

$$= \text{Solun of } \left(\frac{1}{(E - 2)} \right)$$

$$\text{PS} = \text{Solun of } \left(\frac{(1)^k}{(E - 2)} \right)$$

base $b = 1$
it is not
a ch. root

$$\therefore \text{PS} = \frac{(1)^k}{\phi(1)} = \frac{(1)^k}{(1 - 2)}$$

$$\boxed{\text{P.S.} = -1}$$

Complete Soluⁿ

$$a_k = CF + P.S.$$

$$= C_1 \cdot (2)^k + (-1)$$

$$a_k = C_1 \cdot 2^k - 1$$

We know $a_k = T(3^k)$

$$\therefore T(3^k) = C_1 \cdot (2)^k - 1$$

$$T(1) = 1$$

$$3^k = 1 \text{ at } k = 0$$

$$\therefore T(1) = T(3^0) = C_1 \cdot (2)^0 - 1$$

$$1 = C_1 \cdot -1$$

$$C_1 = 2$$

$$\begin{aligned} \therefore T(3^k) &= 2 \cdot (2)^k - 1 \\ &= 2^{k+1} - 1 \end{aligned}$$

* Q:- Find the solution of recurrence relation

$$a_n^2 - 2 \cdot a_{n-1}^2 = 1 \quad \boxed{a_1 = 2}$$

it is not a linear recurrence relation,
but we can convert it into linear recurrence relation

by replacing $a_n^2 = x_n$

∴ Eqⁿ becomes

$$x_n - 2 \cdot x_{n-1} = 1$$

$$x_n - 2x_{n-1} = 1$$

$$\phi(E) = E - 2$$

$$F(n) = 1$$

$$\text{Ch. root} \Rightarrow \boxed{t=2}$$

$$\text{CF} = C_1 \cdot (2)^n$$

$$\text{P.S.} = \text{Soln} \left(\frac{F(n)}{\phi(E)} \right)$$

$$= \text{Soln} \left(\frac{(1)^n}{(E-2)} \right)$$

'1' is not a Ch root

$$\therefore \text{P.S.} = \frac{(1)^n}{\phi(1)} = \frac{(1)^n}{(1-2)}$$

$$\boxed{\text{P.S.} = -1}$$

$\therefore$ Complete Soln

$$x_n = \text{CF} + \text{P.S.}$$

$$x_n = C_1 \cdot 2^n + (-1)$$

$$x_n = C_1 \cdot 2^n - 1$$

$\Downarrow$

$$Q_n^2 = C_1 \cdot 2^n - 1$$

$$Q_n = \pm \sqrt{C_1 \cdot 2^n - 1}$$

$$Q_1 = 2$$

$$\therefore (Q_1)^2 = (2)^2 = C_1 \cdot 2^1 - 1$$

$$\boxed{C_1 = 5/2}$$

$$\therefore Q_n = \pm \sqrt{5 \cdot 2^{n-1} - 1}$$

Q:- Find the solution of recurrence relation

$$\sqrt{a_n} - \sqrt{a_{n-1}} - 2\sqrt{a_{n-2}} = 0$$

$$\boxed{\begin{matrix} a_0 = 1 \\ a_1 = 1 \end{matrix}}$$

Replace $\sqrt{a_n}$ by x_n

$\therefore$ Eqⁿ becomes

$$x_n - x_{n-1} - 2x_{n-2} = 0 \quad \text{P.S} = 0$$

$$\text{Ch. eq}^n: t^2 - t - 2 = 0$$

$$\Rightarrow (t-2)(t+1) = 0$$

$$\text{Ch. root} = t_1 = 2, t_2 = -1$$

$$\text{Cf} = C_1 \cdot (2)^n + C_2 \cdot (-1)^n$$

Complete Soluⁿ

$$x_n = \text{CF} + \text{PS} \rightarrow 0$$

$$x_n = C_1 \cdot (2)^n + C_2 \cdot (-1)^n$$

$\Downarrow$

$$\sqrt{a_n} = C_1 \cdot (2)^n + C_2 \cdot (-1)^n$$

$$a_n = \left(C_1 \cdot (2)^n + C_2 \cdot (-1)^n \right)^2$$

Q:- Find the solution of recurrence relation

$$\sqrt{a_n} - \sqrt{a_{n-1}} - 2\sqrt{a_{n-2}} = 0$$

$$\boxed{a_0 = 1}$$
$$\boxed{a_1 = 1}$$

Complete Soluⁿ

$$x_n = CF + \cancel{PS} \rightarrow 0$$

$$x_n = C_1 \cdot (2)^n + C_2 \cdot (-1)^n$$

$\Downarrow$

$$\sqrt{a_n} = C_1 \cdot (2)^n + C_2 \cdot (-1)^n$$

$$a_n = \left(C_1 \cdot (2)^n + C_2 \cdot (-1)^n \right)^2$$

$$\sqrt{a_0} = \sqrt{1} = C_1 \cdot 2^0 + C_2 \cdot (-1)^0$$

$$\boxed{1 = C_1 + C_2}$$

$$\sqrt{a_1} = \sqrt{1} = C_1 \cdot (2)^1 + C_2 \cdot (-1)^1$$

$$\boxed{1 = 2C_1 - C_2}$$

$$\boxed{C_1 = 2/3}$$

$$\boxed{C_2 = 1/3}$$

Hence,

$$a_n = \left(\frac{2}{3}(2)^n + \frac{1}{3}(-1)^n \right)^2$$

Divide & Conquer Relation

Divide & Conquer Relation

A relation of the type,

$$T(n) = C \cdot T\left(\frac{n}{d}\right) + f(n)$$

is called divide & conquer relation

C & d are constants

We can convert it into a linear recurrence relation by replacing n by d^k

Put $n = d^k$

$\therefore$ Eqn becomes

$$T(d^k) = C \cdot T\left(\frac{d^k}{d}\right) + g(k)$$

function "after parsing d^k " as parameter

$$T(d^k) = C \cdot T(d^{k-1}) + g(k)$$

Put $T(d^k) = x_k$

$\therefore$ Eqn becomes

$$x_k = C \cdot x_{k-1} + g(k)$$

it is a linear recurrence relⁿ.

Q:- Find the solution of the recurrence relation

where $T(1) = 2$

$$T(n) = 7 \cdot T\left(\frac{n}{3}\right) + 2n$$

Put $n = 3^k$

$$\therefore T(3^k) = 7 \cdot T(3^{k-1}) + 2 \cdot 3^k$$

$$T(3^k) - 7 \cdot T(3^{k-1}) = 2 \cdot 3^k$$

$$\text{let } T(3^k) = x_k$$

$\therefore$ Eqn becomes

$$x_k - 7 \cdot x_{k-1} = 2 \cdot 3^k$$

$$\phi(E) = (E - 7)$$

$$F(n) = 2 \cdot 3^{k+1}$$

On Replacing 'k' by 'k+1'

$$\text{Ch. eqn} \Rightarrow t - 7 = 0$$

$$\text{Ch. root} \Rightarrow t = 7$$

$$\text{CF} = C_1 \cdot (7)^k$$

$$\text{P.S} = \text{Soln of } \left(\frac{F(n)}{\phi(E)} \right)$$

$$= \text{Soln of } \left(\frac{2 \cdot 3^{k+1}}{(E - 7)} \right)$$

$$\text{P.S} = 6 * \text{Soln of } \left(\frac{3^k}{(E - 7)} \right)$$

base 'b' = 3

it is not a Ch. root

$$= 6 * \left(\frac{(3)^k}{(3 - 7)} \right)$$

$$\text{P.S} = -\frac{3}{2} \cdot (3)^k$$

∴ Complete Soluⁿ

$$x_k = CF + P.S.$$

$$\Downarrow = C_1 \cdot (7)^k + \left(-\frac{3}{2} \cdot (3)^k\right)$$

$$T(3^k) = C_1 \cdot (7)^k - \frac{3}{2} (3)^k$$

We know $T(1) = 2$

$3^k = 1$ at $k=0$

∴ $T(1) = C_1 \cdot (7)^0 - \frac{3}{2} \cdot (3)^0$

$$T(1) = 2 = C_1 - \frac{3}{2}$$

$$\boxed{C_1 = \frac{7}{2}}$$

$$T(3^k) = \frac{7}{2} \cdot (7)^k - \frac{3}{2} (3)^k$$

We know $3^k = n$

∴ $k = \log_3 n$

$$\therefore T(n) = \frac{7}{2} \cdot (7)^{\log_3 n} - \frac{3}{2} (n)$$

$$\boxed{T(n) = \frac{7}{2} (7)^{\log_3 n} - \frac{3n}{2}}$$

Q. Find the soln of recurrence relⁿ

$$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + (n+1)$$

Where $T(1) = 0$

Put $n = 2^k$

$$T(2^k) = 2 \cdot T(2^{k-1}) + (2^k + 1)$$

Let $T(2^k) = x_k$

Eqⁿ becomes

$$x_k - 2x_{k-1} = 2^k + 1$$

$$\phi(E) = (E - 2)$$

$$F(n) = 2^{(k+1)} + 1$$

Ch eqⁿ $\Rightarrow t - 2 = 0$

Ch. root $\Rightarrow t = 2$

$$CF = C_1 \cdot (2)^k$$

$$P.S = \text{Soln}_{\phi} \left(\frac{F(n)}{\phi(E)} \right)$$

$$= \text{Soln}_{\phi} \left(\frac{2^{k+1} + 1}{(E - 2)} \right)$$

$$P.S = \text{Soln}_{\phi} \left(\frac{2^{k+1}}{(E - 2)} \right) + \text{Soln}_{\phi} \left(\frac{(1)^k}{(E - 2)} \right)$$

base $b = 2$ base $b = 1$

'2' is a root
with multiplicity = 1

'1' is
not
a root

$$P.S = 2 \cdot \left\{ \text{Soln}_{\phi} \left(\frac{(2)^k}{(E - 2)^{1+1}} \right) \right\} + \text{Soln}_{\phi} \left(\frac{(1)^k}{(E - 2)} \right)$$

$\psi(E)$

$$= 2 \cdot \left\{ \frac{1}{\psi(2)} \cdot K C_1 (2)^{k-1} \right\} + \frac{(1)^k}{\phi(1)}$$

$\psi(2) \rightarrow 1$

$$= 2 \cdot \left\{ \frac{1}{1} \cdot K \cdot (2)^{k-1} \right\} + \frac{1}{(1 - 2)}$$

$$P.S = K \cdot 2^k - 1$$

Complete Soluⁿ

$$x_k = CF + P.S.$$

$$\Downarrow = C_1 \cdot (2)^k + (k \cdot (2)^k - 1)$$

$$T(2^k) = C_1 \cdot (2)^k + k \cdot (2)^k - 1$$

We know $T(1) = 0$

$2^k = 1$ at $k = 0$

$$\therefore T(1) = T(2^0) = C_1 \cdot (2)^0 + 0 \cdot (2)^0 - 1$$

$$0 = C_1 - 1$$

$$\boxed{C_1 = 1}$$

$$T(2^k) = 1 \cdot 2^k + k \cdot (2)^k - 1$$

We know $2^k = n$

$$\therefore k = \log_2 n$$

Hence

$$T(n) = n + (\log_2 n) \cdot n - 1$$

$$\boxed{T(n) = n + n(\log_2 n) - 1}$$

GATE 2024

Q:- Let $T(n)$ be the recurrence relation defined as,

$$T(n) = 5 \cdot T(n-1) - 6 \cdot T(n-2) \quad \text{Where } T(0)=1 \text{ \& } T(1)=2$$

Which of the following statement is true

(a) $T(n) = O(n \cdot 2^n)$

✓ (b) $T(n) = O(2^n)$

(c) $T(n) = O(3^n)$

(d) $T(n) = O(n \cdot 3^n)$

Let $T(n) = a_n$

$$a_n = 5a_{n-1} - 6a_{n-2}$$

$$a_n - 5a_{n-1} + 6a_{n-2} = 0$$

$$\phi(E) = E^2 - 5E + 6$$

$$F(n) = 0$$

Ch. eqn $t^2 - 5t + 6 = 0$

Ch root $\Rightarrow t_1 = 3 \text{ \& } t_2 = 2$

$$CF = C_1 \cdot (3)^n + C_2 (2)^n$$

$$PS = 0$$

$\therefore$ Complete Soluⁿ

$$a_n = CF + \cancel{PS} \rightarrow 0$$

$$T(n) = C_1(3)^n + C_2(2)^n$$

$$T(0) = 1 = C_1 \cdot (3)^0 + C_2(2)^0 \Rightarrow \boxed{C_1 + C_2 = 1}$$

$$T(1) = 2 = 3 \cdot C_1 + 2C_2 \Rightarrow \boxed{C_1 = 0}$$

$$\therefore T(n) = 0(3)^n + 1 \cdot (2)^n \quad \boxed{C_2 = 1}$$

$$\boxed{T(n) = (2)^n}$$



Topic : Method of undetermined coefficient

- Method of undetermined coefficient can be used to identify the solution of non-homogeneous part of linear recurrence relation when function $f(n)$ is neither a constant nor an exponential function

ie. particular solution

When

$$f(n) \neq G_1(1)^n$$

$$f(n) \neq G_2 \cdot (b)^n$$

Constant ante

Method of undetermined Coefficient

Consider the following linear recurrence relation

$$\lambda_0 \cdot a_n + \lambda_1 \cdot a_{n-1} + \lambda_2 \cdot a_{n-2} + \dots + \lambda_k \cdot a_{n-k} = f(n) \text{ --- eqn ①}$$

after replacing 'n-k' by 'n', and after using shift operator
Eqⁿ becomes.

$$\lambda_0 \cdot E^k \cdot a_n + \lambda_1 \cdot E^{k-1} \cdot a_n + \lambda_2 \cdot E^{k-2} \cdot a_n + \dots + \lambda_k \cdot a_n = F(n)$$

after replacing
'n' by 'n+k'

$$\text{i.e. } a_n (\lambda_0 \cdot E^k + \lambda_1 \cdot E^{k-1} + \lambda_2 \cdot E^{k-2} + \dots + \lambda_k) = F(n)$$

$$\text{Where } \phi(E) = \lambda_0 \cdot E^k + \lambda_1 \cdot E^{k-1} + \lambda_2 \cdot E^{k-2} + \dots + \lambda_k$$

$$\therefore \text{Eq}^n \text{ becomes } a_n \cdot \phi(E) = F(n)$$

Where $\phi(t) = 0$ is the ch. eqⁿ
And roots of that eqⁿ are called Ch. roots.
Based on Ch. roots we will be able to
define the Complementary function

Method of undetermined Coefficient

Rules for writing Particular Solution

Function $f(n)$ from Eqⁿ-①

$\phi(E)$

Particular Solution

① When function $f(n)$ is of the form,

$$(b)^n \{ C_0 n^S + C_1 n^{S-1} + \dots + C_S \}$$

Exponential
function
with base = b

Polynomial of
degree = S

$C_0, C_1, C_2, \dots$ etc
are known constants

Case ① When $\phi(b) = 0$,
and ' b ' is a
Characteristic root
with multiplicity = m

$(b)^n (A_0 n^S + A_1 n^{S-1} + \dots + A_S) (n)^m$
where $A_0, A_1, \dots, A_S$
are unknown coefficient
and they will be determined by
satisfying the equation

Case ②, When $\phi(b) \neq 0$
i.e., ' b ' is not a
Characteristic root.
{ i.e., ' b ' is ch. root
with multiplicity = 0 }

$(b)^n (A_0 n^S + A_1 n^{S-1} + \dots + A_S) (n)^0$
 $\equiv (b)^n (A_0 n^S + A_1 n^{S-1} + \dots + A_S)$
where $A_0, A_1, \dots, A_S$
are unknown coefficient

Method of undetermined Coefficient

Rules for writing Particular Solution

Function $f(n)$ from Eqⁿ - ①

$\phi(E)$

Particular Solution

② When function $f(n)$ is of the form,

$$\{C_0 n^S + C_1 n^{S-1} + \dots + C_S\}$$

Polynomial of degree = S

$C_0, C_1, C_2, \dots$ etc are known constants

$$\equiv (1)^n \{C_0 n^S + C_1 n^{S-1} + \dots + C_S\}$$

Case ① When $\phi(1) = 0$ and '1' is a Characteristic root with multiplicity = m

Case ② When $\phi(1) \neq 0$ i.e., '1' is not a Characteristic root. i.e., '1' is Ch. root with multiplicity = 0

$$\cancel{(1)^n}^1 (A_0 n^S + A_1 n^{S-1} + \dots + A_S) \cdot (n)^m$$

where $A_0, A_1, \dots, A_S$ are unknown coefficient and they will be determined by satisfying the equation

$$\cancel{(1)^n}^1 (A_0 n^S + A_1 n^{S-1} + \dots + A_S) \cdot (n)^0 \equiv \cancel{(1)^n}^1 (A_0 n^S + A_1 n^{S-1} + \dots + A_S)$$

where $A_0, A_1, \dots, A_S$ are unknown coefficient

#Q. Find the solution of recurrence relation

$$a_n - 3a_{n-1} = n+3$$

Put " $n-1$ " = n & use shift operator

$$E \cdot a_n - 3a_n = (n+1)+3$$

$$a_n(E-3) = n+4$$

$$\phi(E) = E-3$$

Chr root $\Rightarrow t=3$

$$\therefore CF = C_1 \cdot (3)^n$$

$f(n)$

$$f(n) = (n+3)$$

$$= (1)^n (n+3)$$

base $b=1$

Polynomial of deg = 1

'1' is not a characteristic root { i.e. root with multiplicity = 0 }

$\therefore$ Particular Soluⁿ = $(1)^n (An+B) \cdot (n)^0$

Polynomial of deg = 1
& A & B are unknown constants

$$a_n^{(p)} = An+B$$

We have, $Q_n^{(P)} = An + B$

We know, Particular solution must satisfy the given linear recurrence relⁿ

given recurrence relⁿ

$$a_n - 3a_{n-1} = (n+3)$$

∴ $Q_n^{(P)} - 3 \cdot Q_{n-1}^{(P)} = n+3$

ie $(An+B) - 3 \cdot (A(n-1)+B) = n+3$

Put $n=0 \Rightarrow (A \cdot 0 + B) - 3(A(0-1) + B) = 0 + 3 \Rightarrow 3A - 2B = 3$ } $A = -1/2$

Put $n=1 \Rightarrow (A \cdot 1 + B) - 3(A(1-1) + B) = 1 + 3 \Rightarrow A - 2B = 4$ } $B = -9/4$

∴ Particular Solution $= \left(-\frac{1}{2}n + \left(-\frac{9}{4} \right) \right) \Rightarrow Q_n^{(P)} = -\frac{n}{2} - \frac{9}{4}$

Complete Solution

$$a_n = CF + P.S.$$

$$a_n = C_1 \cdot (3)^n + \left(\frac{-n}{2} - \frac{9}{4} \right)$$

← using boundary Condⁿ
we will be able to
identify constant "C₁"

H.W.



#Q. Find the solution of recurrence relation

$$a_n - a_{n-1} = n+2$$

$$\phi(E) = E - 1$$

$$\text{Char root} \Rightarrow t = 1$$

↓

$CF = C_1(1)^n$

$$f(n) = (n+2)$$

$$= (1)^n \cdot (n+2)$$

↑ '1' is a root with multiplicity = '1'

∴ Particular Solution = $(1)^n (An+B)n^1$

$P.S = n \cdot (An+B)$

#Q. Find the solution of recurrence relation

$$a_n - 2a_{n-1} + a_{n-2} = 3n+5$$

$$\phi(E) = E^2 - 2E + 1$$

$$\text{Ch. eqn} \Rightarrow t^2 - 2t + 1 = 0$$

$$\text{Ch. root} \Rightarrow \underbrace{t_1=1, t_2=1}_{\text{two equal roots}}$$

$$\begin{aligned} \text{C.F} &= (C_1 + C_2 n) \cdot (1)^n \\ &= C_1 + C_2 \cdot n \end{aligned}$$

$$f(n) = 3n + B$$

$$= (1)^n \cdot (3n + B)$$

'1' is a root with multiplicity=2

$$\therefore \text{P.S} = (1)^n \cdot (An + B) \cdot n^2$$

$$\boxed{\text{P.S} = n^2 \cdot (An + B)}$$

#Q. Find the solution of recurrence relation

$$a_n - a_{n-1} = 2^n \cdot n$$

Ch. root $\Rightarrow \boxed{t=1}$

$$f(n) = 2^n \cdot n$$

$$= 2^n (1 \cdot n + 0)$$

Polynomial of deg=1
 $\hookrightarrow$ '2' is root with multiplicity=0

$$\therefore \text{P.S.} = (2)^n \cdot (An+B) \cdot n^0$$

$$\boxed{\text{P.S.} = 2^n \cdot (An+B)}$$

#Q. Find the solution of recurrence relation

$$a_n - 3.a_{n-1} = 3^n(n+2)$$

Ch. root $\Rightarrow t=3$

$$f(n) = 3^n(n+2)$$

'3' is a root with multiplicity = 1

$$\therefore P.S = (3)^n \cdot (An+B) \cdot n^1$$

$$P.S = n \cdot 3^n \cdot (An+B)$$



2 mins Summary



✓ **Topic**

Solution of recurrence relation using method of characteristic roots

✓ **Topic**

Method of undetermined coefficient for particular solution

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THANK - YOU